PERCEPTRON HACKATHON 2026 · ROUND 1 ABSTRACT · THEME 2 — CAUSE AND EFFECT IN AI DECISIONS · PROBLEM 3

MIRAGE: Did the New Quoting Strategy Make Money, or Did It Just Look Like It?

Counterfactual evaluation of market-making policies, with uncertainty and sensitivity to hidden informed flow

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1. Problem & Theme

We chose Problem 3 and set it inside a market maker's quoting desk, where "did the new strategy work?" is asked daily and often answered wrongly. A quoting policy decides which trades it receives: tight quotes win fills, including from informed traders who pick it off. So the policy shapes the very PnL used to judge it: exactly this problem's trap.

2. Understanding of the Challenge

A before/after number fails for three reasons. **(i) Regime confounding:** if the new policy launched in a calm week, the PnL jump is volatility, not strategy. **(ii) Support:** production quoting is near-deterministic, so logs barely contain the actions a new policy would take; importance weights explode and estimates become extrapolation. **(iii) Hidden confounding:** desks react to information that is never logged (e.g. a sense that flow is toxic). Logged propensities are then wrong and no estimator is unbiased; the honest answer is a bound, not a number. A credible system must estimate, quantify uncertainty, and state how much hidden bias would overturn its conclusion.

3. Proposed Approach

- **Simulator with ground truth.** A Python/Numba limit-order-book simulator with Poisson noise traders, a two-state Markov volatility regime, and Glosten–Milgrom-style informed traders holding a private signal U about the future mid-price. Our market maker picks one of 9 actions per decision (half-spread {1,2,3} ticks × inventory skew {−1,0,+1}) from context x = (realised volatility, book imbalance, inventory, recent signed flow, spread). Because we own the simulator, any policy's true value is obtained by replaying it on identical random seeds.
- **Logged data.** Production policy π_0 is a softmax (temperature τ) over a rule-based score and also sees a noisy copy of U with strength β, but only $(x, a, \pi_0(a|x), r)$ is logged. Reward r is the h-step markout PnL of any fill minus an inventory penalty λI^2, the standard desk metric, keeping each decision close to a contextual bandit.
- **Estimator bank.** Via Open Bandit Pipeline (obp): IPS, SNIPS, Direct Method (LightGBM reward model), cross-fitted Doubly Robust, Switch-DR and DRos, against two naive baselines (before/after PnL; replay-matching).
- **Uncertainty → decision.** Episode-level block-bootstrap CIs and an empirical-Bernstein high-confidence lower bound (HCOPE). The system says **DEPLOY** only if the lower bound on $V(\pi_1) - V(\pi_0)$ is positive.
- **Sensitivity.** A marginal sensitivity model (true vs. logged propensity odds within $[1/\Gamma, \Gamma]$) with sharp quantile-balancing bounds, reporting the breakdown Γ^* at which the verdict flips.
- **Policy Court.** Streamlit dashboard comparing naive, every estimator and the truth; drag Γ and watch the verdict flip.

4. Minimum Viable

(a) Standard OPE: IPS and DR running on simulator logs and on the Open Bandit Dataset. **(b) Concrete divergence:** a scripted "calm-week launch" where naive before/after shows a large gain while both the true value and DR show a loss (already reproduced, Fig. A).

5. Stretch Ambitions

Both listed goals: CIs plus HCOPE (uncertainty), then Γ-sensitivity bounds (unobserved confounding). Beyond them, a support stress test sweeping τ to map where each estimator breaks.

6. Novelty / Differentiation

- To our knowledge, no published OPE work targets market-making quote policies with markout rewards; we bring causal evaluation to the sponsor's core decision.
- **Validated sensitivity analysis.** Sensitivity bounds are normally unverifiable. Our simulator dials the true confounding strength β, so we compute the actual Γ and test whether the bounds really contain the truth.
- We output a decision with a guarantee, not a point estimate.

7. Evaluation Plan

Ground truth from simulator replay ($\geq 10^5$ decisions per policy, 30 seeds) across 50 candidate policies. Metrics: relative error vs. true value, sign accuracy of uplift, CI coverage and width, effective sample size, Γ-bound coverage at the true

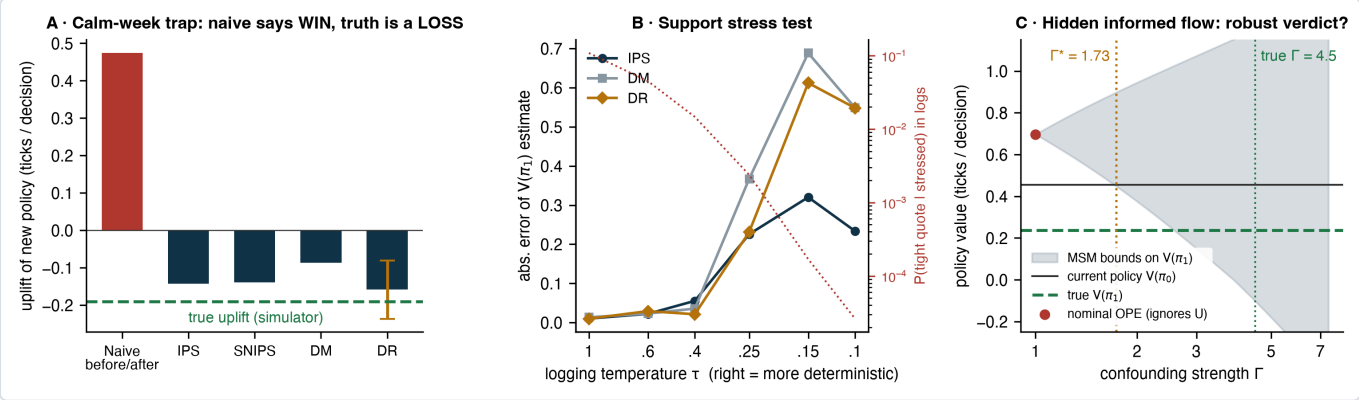
Γ, and deploy-decision accuracy and regret vs. naive A/B. Real-data check: on the Open Bandit Dataset, estimate the Bernoulli Thompson Sampling policy's CTR from uniform-random logs and compare with its on-policy CTR (Saito et al., 2021 protocol). Simulator realism: fat-tailed returns, volatility clustering, negative informed-fill markouts.

8. Feasibility & Risks

Already de-risked. A working prototype (github.com/ac265640/mirage-ope; 3 spread actions, no order book yet) reproduces the core claims across 10 independent seeds: naive before/after gets the sign wrong 10/10, DR gets it right 10/10 with its 95% CI covering the truth 10/10, and mean error drops from 0.63 to 0.03 ticks/decision. Under hidden confounding, nominal OPE wrongly says "deploy" 10/10, while MSM bounds contain the truth at the true Γ 10/10 and flag the verdict as fragile (Γ* = 1.4–2.3).

CPU-only Python with open libraries. **Roles:** Amit — simulator, data pipeline and Policy Court dashboard (built end-to-end Python/Pandas, PostgreSQL and Power BI analytics in his CustomerLens project); Anandita — estimator bank and Open Bandit validation; sensitivity analysis shared. **Milestones:** Wk1 simulator + truth oracle; Wk2 estimators + calm-week case (MVP); Wk3 CIs, HCOPE, OBD validation (midway report, 1 Dec); Wk4 confounding knob + Γ-bounds; Wk5 50-policy benchmark, dashboard, rehearsal.

Risks: simulator judged unrealistic → stylised-fact calibration plus the real-data OBD check; inventory carry-over breaks the bandit assumption → short-horizon markouts with inventory in context, measuring residual bias against full-episode truth.



Prototype results (toy world, seed 7). A: naive +0.47 ticks/decision vs. truth −0.19; DR −0.16 [95% CI −0.24, −0.08]. B: as π_0 turns deterministic, tight quotes vanish from stressed-market logs and every estimator fails. C: nominal OPE says π_1 wins; the bounds overturn that at Γ* = 1.73 and contain the truth at the true Γ.

Key references: Saito et al., Open Bandit Dataset & Pipeline (NeurIPS D&B 2021) · Dudík et al., Doubly Robust Policy Evaluation (ICML 2011) · Su et al., DR with Optimistic Shrinkage (ICML 2020) · Thomas et al., High-Confidence Off-Policy Evaluation (AAAI 2015) · Kallus & Zhou, Confounding-Robust Policy Improvement (NeurIPS 2018) · Dorn & Guo, Sharp Sensitivity Analysis for IPW via Quantile Balancing (JASA 2023) · Glosten & Milgrom (JFE 1985).

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MIRAGE: system architecture

The left column is the world the desk lives in. The right column is what only a simulator can give us: the truth.

